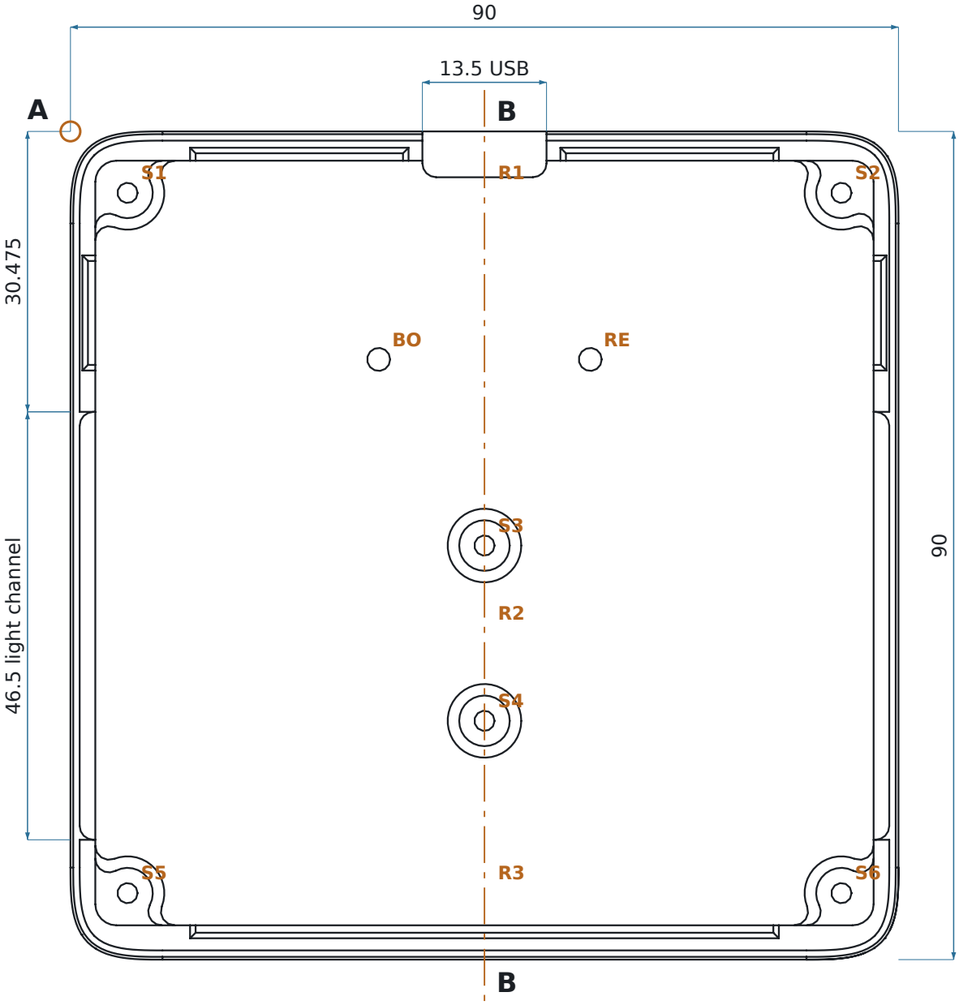
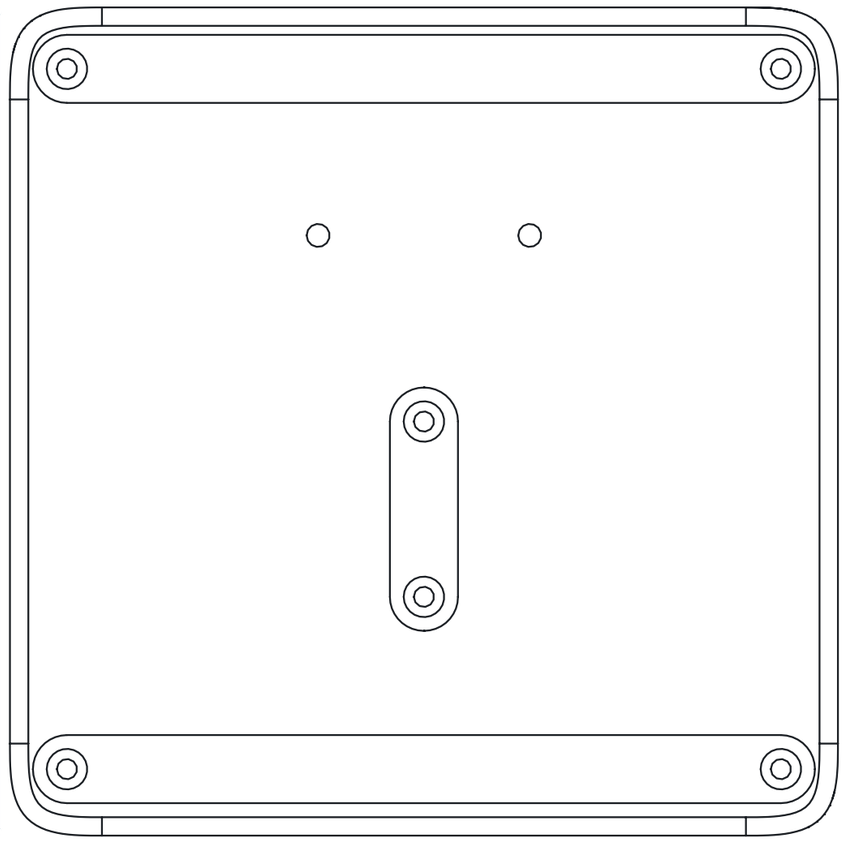
FROM ABOVE



FROM BELOW (mirrored left to right)

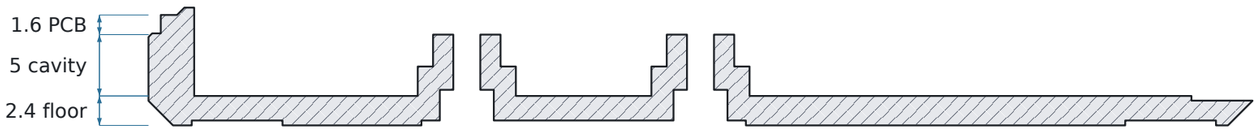


macropad case: base, CNC variant

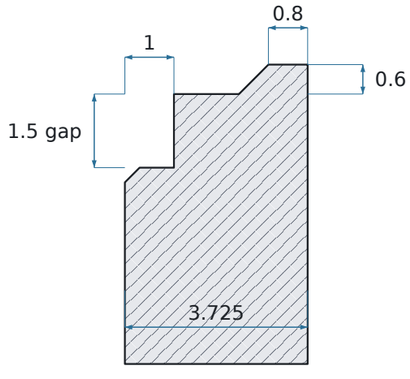
Holes and recesses, from datum A

tag	X	Y
⌀2.2 through; counterbore ⌀4.4 to 2.9 from below		
S1	6.2	6.675
S2	83.8	6.675
S3	45	45
S4	45	64.05
S5	6.2	82.775
S6	83.8	82.775
⌀2.5 through (BOOT)		
BO	33.5	24.775
⌀2.5 through (RESET)		
RE	56.5	24.775
slot 85 x 7.4, full-radius ends, 0.4 deep from below		
R1	45	6.675
slot 7.4 x 26.45, full-radius ends, 0.4 deep from below		
R2	45	54.525
slot 85 x 7.4, full-radius ends, 0.4 deep from below		
R3	45	82.775

SECTION B-B



GAP AND LAP JOINT, front wall, 8:1



Scale 1.5:1 plans, 2:1 section, 8:1 detail, on A3. Dimensions in mm, computed from the model.
Datum A: back-left corner of the box round the outline, seen from above. X to the right, Y toward the front.
3-axis, two setups: inside up, then underside up.
Every pocket corner is R1.5 or larger: a 3 mm end mill reaches them all.
45 degree chamfers: 2 on the underside rim, 0.3 on the outer skin's top edge at the gap, the flank and ends of the lap ridges, the USB lead-in.
The rim inside the gap stays sharp: the top seats on it.
The rim steps in 1 mm over the top 1.5 mm, all round: that is the gap. Cut it with the side of the end mill in setup 1.
Screw holes go straight through: M2 x 8 socket head from below.
Outline corners: one quintic Bezier each, control points 10, 6 and 2 mm from the box corner on both sides. Take the curve from the STEP file.
General tolerance: ISO 2768-m is a proposal. Nothing has been made or measured.